

SmartHire AI — Database Schema (v1.0)

Single-table design, matching the PRD Section 8 data model exactly. No user accounts (auth is out of scope).

Table: `candidate_analyses`

sql

```
CREATE TABLE candidate_analyses (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  candidate_name VARCHAR(150) NOT NULL DEFAULT 'Unknown Candidate',  
  resume_file_path VARCHAR(255) NULL,  
  resume_text LONGTEXT NOT NULL,  
  ats_score INT NOT NULL CHECK (ats_score BETWEEN 0 AND 100),  
  ats_breakdown JSON NOT NULL,  
  resume_summary TEXT NULL,  
  strengths JSON NULL,  
  improvement_suggestions JSON NOT NULL,  
  interview_questions JSON NOT NULL,  
  interview_readiness_score INT NOT NULL CHECK (interview_readiness_score BET  
  recommended_roles JSON NULL,  
  analysis_date DATETIME NOT NULL,  
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);  
  
CREATE INDEX idx_analysis_date ON candidate_analyses (analysis_date DESC);
```

Field Notes

Field	Notes
<code>candidate_name</code>	Extracted from resume text if possible; falls back to a default label
<code>resume_file_path</code>	NULL when the paste-text fallback was used (no file exists)
<code>resume_text</code>	Full extracted/pasted text — used as AI context and for re-analysis if needed
<code>ats_breakdown</code>	JSON: <code>{ formatting, keywords, structure, clarity }</code> sub-scores
<code>strengths</code>	JSON array of short strength statements
<code>improvement_suggestions</code>	JSON array: <code>[{ area, issue, recommendation }]</code>
<code>interview_questions</code>	JSON array: <code>[{ category, question, basedOn }]</code>
<code>recommended_roles</code>	Nullable — only populated if the v1.1/stretch feature is built

Constraints

- `ats_score` and `interview_readiness_score` are constrained to 0–100 at the DB level as a safety net against malformed AI output.
- `resume_text`, `ats_breakdown`, `improvement_suggestions`, `interview_questions` are `NOT NULL` — these are the Must-Have fields; a record without them is not a valid analysis.
- No foreign keys — single-table design, no relationships needed since there are no user/account tables in v1.0.

Validation Against PRD User Stories

User Story	Supported By
US-01 Upload PDF/DOCX	resume_file_path, resume_text
US-02 ATS score	ats_score, ats_breakdown
US-03 Improvement suggestions	improvement_suggestions
US-04 Interview questions	interview_questions
US-05 Readiness score	interview_readiness_score
US-06 Recruiter candidate list	All fields queryable via <u>SELECT id, candidate_name, ats_score, interview_readiness_score, analysis_date</u>
US-07 Recruiter full report	<u>SELECT *</u> by <u>id</u> returns everything needed

Every user story from the PRD is fully covered by this single table — no missing fields, no unnecessary ones.